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Interpretable Rules for Rare Build-Up Failures in Football (Soccer): Sequential Pattern Mining Meets Inductive Logic Programming

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Abstract. In football, rare but costly build-up failures are hard to detect in routine video review because any given failure pattern may occur only a few times per season. Existing approaches assign numerical values to actions or use black-box models requiring many examples; neither exposes the temporally ordered sequence in which a build-up breaks down. We present a hybrid method that learns short, temporally ordered rules from a single team-season, with strict within-fold separation of training and test data and coach review of prioritized rules. It converts StatsBomb event and 360 freeze-frame data into Prolog facts, mines gap-constrained ordered patterns within each training fold, and applies Popper inductive logic programming with minimum description length to learn short, auditable rules for sequences ending in an unrecovered loss of possession in a team's own half. On Barcelona 2020/21, the hybrid reaches 25.3% precision against a 13.4% base rate (lift 1.89; 95% match-cluster bootstrap CI 1.48–2.33); a context-only logistic baseline using sequence length and start coordinates has lift 1.32 (1.09–1.54). On Bayer Leverkusen 2023/24, hybrid lift is 1.26 (0.63–1.90) versus 1.48 (1.31–1.67) for the baseline; neither comparison supports a stable winner across both team-seasons, suggesting the advantage of ordered rules may be team-specific. The precision-oriented operating point (rule-arm recall 6–17% vs 55–60% for the baseline) suits hypothesis generation, where false positives are costly for expert review. An AFC-certified coach approved 7 of 12 reviewed rules, including four conditionally. The system generates testable hypotheses for video review, not predictions; lack of full tracking limits spatial accuracy.

Keywords— football build-up, inductive logic programming, sequential pattern mining, interpretability, StatsBomb 360

1. Introduction

In build-up —the phase in which a football team advances the ball from its own half toward the opponent's goal—losing possession is particularly concerning for coaches: defenders have little time to reorganize and the opponent is close to goal. The number of players behind the ball has been reported as an important factor in counterattacks after a turnover [1]. Frequent patterns are familiar to coaches, but rare multi-action configurations—a two- or three-action sequence seen only a few times in a season—can escape both memory and routine video review, even though one occurrence may lead to a goal.

Common approaches assign numerical value to each action [2], [3], [4] or use black-box models that require many examples [5]. Such scores indicate how valuable an action is but not why a sequence broke down, and black-box models are hard to audit and unreliable when only a few examples exist. We study a different regime: one team, one season, roughly 175–177 rare failure sequences (negative examples), using publicly available event and 360 freeze-frame data, with an output that can be read and audited. Inductive logic programming (ILP) can learn readable logic programs from limited data [6], but direct search over action sequences is combinatorial. We therefore convert gap-constrained sequential patterns—patterns with bounded gaps between actions—into predicates and use Popper [7], an ILP system that learns programs from failures: each failed candidate rules out similar programs, pruning the hypothesis space. Its minimum description length (MDL) mode handles noisy data [8] and selects a compact subset of the patterns. To our knowledge, this combination has not been reported for build-up failures. Our aim is interpretability, not predictive or causal superiority.

We make four contributions: (1) a hybrid pipeline with strict within-fold separation of training and test data for the primary out-of-fold (OOF) evaluation (Fig. 1); (2) a comparison of end-to-end procedures—the hybrid, the sequential pattern miner alone, pattern-free Popper, classification and regression trees (CART) [9], and a simple sequence-context logistic baseline; (3) modular background knowledge (BK) that analysts can activate or filter without changing the learner; and (4) coach review that also records insights from rejected rules. The remainder of the paper describes the data, method, evaluation, and results.

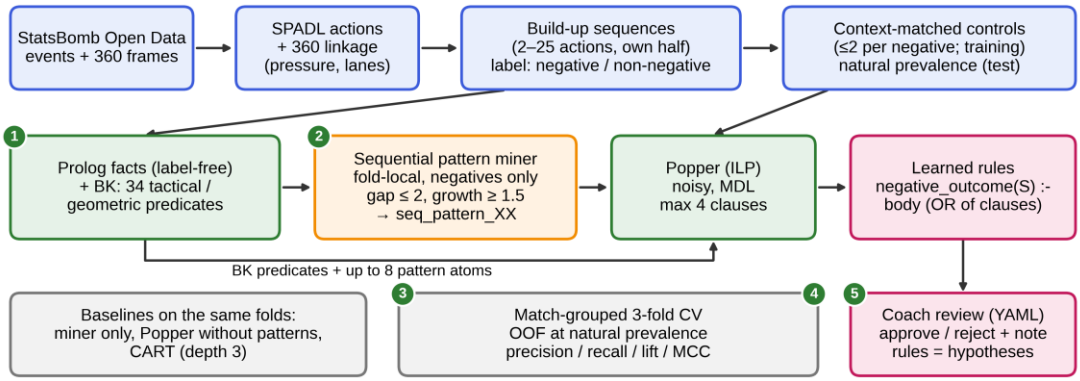


Fig.1.Five safeguards: (1) unlabeled facts; (2) pattern mining inside each fold; (3) match-grouped folds; (4) natural-prevalence evaluation; (5) review of prioritized rules.

2. Related Works

Valuing Actions by Estimating Probabilities (VAEP) measures scoring-probability changes over the next actions [2], while expected threat (xT) values ball movement between pitch zones [3] and only progressive actions [4]. These methods give numerical scores but not the configuration in which a sequence breaks down. Following Rudin [5], we prefer an intrinsically interpretable model.

Decroos et al. [10] discover attacking tactics by clustering phases. Biermann et al. [1] predict counterattacks with logistic regression over comprehensible features. Dash et al. [11] study 2,413 defensive transitions by Barcelona and Real Madrid using spatial indicators and gradient boosting on tracking data. These outputs are clusters, weights, or scores rather than logic rules. Our spatial layer relies on 360 frames that record only players in view [12].

ILP learns logic theories from examples and background knowledge [13]. Popper prunes the hypothesis space by learning from failures [7], uses Clingo for solving [14], and supports MDL programs for noisy data [8]. In sport, Vercruyssen et al. [15] combine qualitative spatial reasoning and ILP for pass prediction, Van Haaren et al. [16] discover attacking patterns with ILP, and Jacobs et al. [17] use ILP to verify agent behavior in RoboCup. None targets build-up failure.

Emerging patterns are itemsets whose support rises sharply between classes [18], making growth rate a natural criterion for patterns that distinguish failed sequences. PrefixSpan [19] and gap-constrained generalizations [20] grow patterns through projected prefixes. Data leakage can produce optimistic results [21], while random cross-validation underestimates error for dependent data [22]; we therefore split by match.

3. Data and Problem Definition

We use StatsBomb Open Data events, line-ups, and 360 frames for two team-seasons (Table I). Coordinates are converted to Soccer Player Action Description Language (SPADL) [2] dimensions (105 × 68 m), with the focal team always attacking toward positive x. A 360 frame records only visible players and does not provide identities for non-actors; continuous tracking is unavailable. For Leverkusen, 31 of 34 available 2023/24 matches were included because only those 31 had usable 360 frames.

Each sequence starts at the first focal-team action in a possession chain in the team's own half (x≤52.5 m), with at most one sequence per possession. Sequences from throw-ins, corners, or free kicks are excluded, except short or medium goal-kick build-ups. A sequence ends at the first shot, failed action, controlled entry into the attacking region (deep quarter, zone 14, or deep half-space; for a pass, same-region receipt must confirm entry), 25-action cap, or end of possession. We retain $2 \le n \le 25$ actions (median length: 8). For $S = (a_1, \dots, a_n)$, the learner sees only prefix $(a_1, \dots, a_{n-1})$:

TABLE I. DATASET

Measure	Barcelona 20/21	Leverkusen 23/24
Matches with 360 data	35	31 of 34
Build-up sequences	1,308	1,086
Negative / non-negative	175 / 1,133	177 / 909
Base rate π	13.38%	16.30%
360 frame coverage	92.7%	85.8%

$$y(S) = 1 \Leftrightarrow x_{\text{loss}}(S) \le 52.5 \ \wedge \ \neg \text{Rec}_5(S) \ \wedge \ \neg \text{Foul}(S) \qquad (1)$$

Here x_{loss} is the turnover location, Rec_5 denotes controlled recovery within five seconds, and Foul denotes a foul as the opponent's first action (an ambiguous boundary case). A non-negative sequence does not necessarily indicate tactical success. The base rate is $\pi = |\{S : y(S) = 1\}| \,/\, |D|$.

For Popper and CART training, we select at most $r = 2$ context-matched controls per negative, matched on start zone, pressure, and sequence length. The miner uses all training sequences at natural prevalence, and evaluation always uses all sequences from held-out matches at natural prevalence. With P negatives, N_0 non-negatives, and sampling fraction $s = rP/N_0$, natural-prevalence precision can be approximated from sampled confidence c as:

$$\text{Prec}_{\text{nat}} \approx \frac{c}{c + (1-c)/s} . \qquad (2)$$

For Barcelona, $s = 0.309$; for Leverkusen, $s = 0.389$. For example, $c = 0.6$ corresponds to approximately 0.32 and 0.37, respectively. Sampled confidence is therefore not directly comparable to natural-prevalence rates.

Off-ball movement and speed are unavailable. We excluded terminal-action-dependent predicates (possible label leakage) and concepts not precisely definable with these data. Predicates such as rest_defence_exposed are proxies for defensive shape, not direct measurements.

4. Method

The five-stage pipeline (Fig. 1) converts data to Prolog facts [23], defines background knowledge, mines sequential patterns [19], [20], learns rules with Popper [7], [8], and sends prioritized candidates for coach review. Facts carry no

labels; labels enter only during learning. We call a sequence ending in an unrecovered loss of possession a failure sequence, and any other a non-failure sequence. The learner's target predicate is negative_outcome(S), referred to in the text as the failure outcome.

4.1 Background Knowledge

Tactical and geometric concepts are declarative Prolog rules over base facts (SWI-Prolog [23]). Coordinates use the 105 × 68 m scale of Section 3. The pitch has three thirds (defensive x≤35, middle, attacking) and five lanes, aggregated into left/centre/right channels for mining. Pressure means at least one opponent within 3 m; an open passing lane means a teammate is reachable with no opponent within 2 m of the lane. Each sequence-level predicate is an existential prefix wrapper: seq_X(S)⇔∃ j≤n-1: X(a_j). The active vocabulary has 34 predicates. This is propositionalizing [24]: relational concepts may be multi-ary in BK, but the learner sees unary sequence-level features to constrain combinatorial growth. It therefore does not directly learn multi-player relations. Table II defines the eight predicates in Table V; the full 34-predicate vocabulary is in the repository.

TABLE II. OPERATIONAL DEFINITIONS OF PREDICATES IN TABLE V

Predicate	Operational definition and caveat
under_pressure	At least one opponent within 3 m (weakest pressure threshold).
rest_defence_exposed	Fewer than two non-goalkeeper teammates ≥5 m behind the carrier; a proxy (in the defensive third, partly reflects the ball being with the deepest player).
dead_end_support	Valid frame and no teammate with an open passing lane.
line_pass	Successful pass from a non-central lane to the same lane, non-lateral direction; includes backward passes and is not necessarily longitudinal.
connecting_pass	Successful short pass (<10 m) that is non-progressive.
first_time_play	Possession duration ≤0.5 s; a temporal proxy.
fast_sequence	Actions per second ≥70th percentile.
flank_backward_carry	Carry from a flank to the same flank with $\Delta x \leq -4$ m.

4.2 Sequential Pattern Mining

Each item is a tuple (predicate, action type, channel, pitch third), where the action type is the StatsBomb event category (e.g., pass or carry), and $M(a)$ is the set of items for action a. A pattern $p = (\iota_1, \ldots, \iota_k)$, with $k \in \{2,3\}$ and maximum gap $g = 2$ [20], matches sequence S if:

$$S \vDash p \Leftrightarrow \quad \exists\, t_1 < \cdots < t_k \leq n-1\colon \forall j \leq k,\ \iota_j \\ \wedge\ \forall j < k,\ 0 \leq t_{j+1} - t_j - 1 \leq g \tag{3}$$

Patterns are mined only from failure training sequences within each fold (closed patterns—those not subsumed by a more general pattern with identical support—under the stated search bounds), then counted on non-failure training sequences. The terminal action is removed before mining because it is the loss of possession itself and would otherwise leak the label. For the same reason, predicates that inspect the following action (e.g., line_pass and first_time_play) are masked on the last retained action. To reduce confounding by context composition (start location, pressure, and length category), rates are standardized over strata S shared by the failure and non-failure classes. . Let T_s and C_s be the negative and non-negative training examples in stratum s, $w_s = |T_s| \,/\, \sum_{s'} |T_{s'}|$, and $\alpha = 0.5$:

$$\begin{aligned} \tilde{r}_X(p) &= \sum_{s \in \mathcal{S}} w_s \, \frac{\sup_{X_s}(p) + \alpha}{|X_s| + 2\alpha},\ X \in \{T, C\} \\ \mathrm{GR}(p) &= \frac{\tilde{r}_T(p)}{\tilde{r}_C(p)} \end{aligned} \tag{4}$$

Here, $\sup_{T_s}(p)$ and $\sup_{C_s}(p)$ denote the support of p, i.e., the number of sequences in T_s and C_s, respectively, that match p, and $\mathrm{GR}(p)$ is the growth rate of p. A pattern is accepted when it matches at least five failure sequences and GR≥1.5 [18]; this is a training-set ranking criterion, not a significance test. Accepted patterns are sorted by GR, then failure support, then shorter length. Up to the top eight become seq_pattern_XX(S), expressing only whether the ordered pattern occurs in the prefix. Our miner is a bounded, prefix-based, gap-constrained, support-closed variant, not a canonical implementation of Gap-BIDE [25].

4.3 Popper Learning

In its noisy setting, Popper searches for a program with minimum MDL cost:

$$\mathrm{cost}(h) = \mathrm{size}(h) + \mathrm{FN}(h) + \mathrm{FP}(h) \tag{5}$$

Size counts total literals across all clauses of h; FN/FP are false negatives (failure training sequences not covered by h) and false positives (non-failure training sequences covered by h). A single-atom clause has size 2 and reduces cost only if it adds at least three net true positives (TP−FP), consistent with low_support=3. Settings: target negative_outcome (S) (the failure outcome); at most two variables, three body literals, and four clauses (logical OR); BK with 34 predicates and at most eight pattern atoms; 600 seconds per fold. With 34 predicates, the unconstrained program space is upper-bounded at roughly 10^{15}, motivating clause and time limits. A timeout or NO SOLUTION is not counted as success.

4.4 Arms and Evaluation Protocol

Five arms share outer match-grouped folds and natural-prevalence test population (13.4% failures for Barcelona): hybrid (34 predicates plus up to eight pattern atoms, Popper clauses OR-combined); miner only (OR over mined patterns, no Popper selection); Popper without patterns; CART; and context-only logistic regression using sequence length (n_actions) and start coordinates (start_x, start_y). Logistic model standardizes these three features, uses C=1.0 (scikit-learn default), and trains on all outer-training sequences at natural prevalence. Threshold selected by max F1 on three-fold match-grouped inner OOF predictions within each outer training set (ties: precision, then higher threshold); outer test fold not used. CART uses Gini splits, max depth 3, min leaf size 10; terminal leaves predict majority class, ties to failure class. Popper and CART train on context-matched controls; miner trains at natural prevalence. Comparisons concern complete procedures, not isolated algorithm effects. Because training populations differ, these ablations do not isolate miner or Popper contributions; controlled comparison with identical populations is future work.

For the reported hybrid runs, the sequential-pattern settings were fixed at minimum growth rate 1.5, minimum support 5, at most eight selected patterns, and smoothing α=0.5; these settings were not tuned against outer test-fold outcomes.

We use three-fold cross-validation (CV) grouped by match with a fixed seed. Predictions are aggregated OOF across all test sequences at natural prevalence:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad \mathrm{Lift} = \frac{P}{\pi} \qquad (6)$$

Here, TP, FP, FN, and TN are the numbers of true positives, false positives, false negatives, and true negatives, where a positive is a failure sequence. Table III also reports the Matthews correlation coefficient (MCC). For dependence among same-match sequences, 95% percentile confidence intervals for OOF lift used 20,000 match-cluster bootstrap replicates (seed 20261019 Barcelona; 20261020 Leverkusen). Matches were resampled with replacement within their original test fold, retaining all evaluated sequences from each sampled match together [26]. Lift (precision divided by outcome base rate) was recomputed every replicate. The same resampled matches supported paired lift differences between methods. Fold-specific fitted models and predictions were held fixed; intervals quantify test-match sampling uncertainty conditional on saved models and split, not full-pipeline refitting uncertainty. Wilson intervals for selected in-sample rule rates in Table V are sequence-level and not match-cluster adjusted [27].

We also report each rule's seasonal rate: $h(p)$ is the number of negative matches and $\mathrm{cov}(p)$ is the total number of matching sequences.

$$\rho(p) = \frac{h(p)}{\mathrm{cov}(p)} \qquad (7)$$

Intermediate candidates were curated using discovery-fold test evidence: ranked by strongest usable test-fold lift, with support as tie-breaker. Prioritized rules were then scored on a separate fixed 75/25 match split; we flag support below 3 and confidence decline exceeding 50% from training to this review holdout. An Asian Football Confederation (AFC)-certified coach, without blinding, marks each rule approved or rejected and records notes. Because discovery-fold test results informed candidate ranking and the review split scored selected rules, candidate-level rule scores and aggregate-model AUCs in Section 5.3 are exploratory estimates influenced by selection, not independent evidence of generalization. This post-hoc curation does not determine primary Table III arm-level OOF predictions, which come from models learned within each outer training fold.

Reproducibility. StatsBomb Open Data is publicly available, but this alone does not identify the exact analysis subset or fold assignments. The accompanying repository provides the code and configuration, `match_split_manifest.csv` (included matches, outer-CV test-fold IDs, coach-review assignments), the original per-match OOF counts for the four rule procedures, and the context-baseline script and outputs (per-match confusion counts, the 20,000-replicate bootstrap intervals, and paired lift differences):

https://github.com/moeinHP-aussie/Interpretable-Rules-for-Rare-BuildUp-Failures-in-Football-by-Pattern-Mining-and-ILP

5. Results

5.1 Arm Comparison

Table III reports the five-method OOF comparison at natural prevalence. Methods share outer folds, sequences, and test labels (SHA-256 verified per fold); the context-baseline threshold is selected within each outer training fold. Arms return different artifacts (motif clauses, pattern-free clauses, tree paths, or a context score); Tables III–IV compare enrichment only, not interpretability or coach utility.

On Barcelona, hybrid lift was 1.89 (95% CI 1.48–2.33), versus 1.32 (1.09–1.54) for the context baseline. In Leverkusen, hybrid lift was 1.26 (0.63–1.90), versus 1.48 (1.31–1.67); the hybrid interval includes 1, so its enrichment is inconclusive. Paired comparisons against all procedures are in Table IV; they do not establish a general ranking and are not adjusted for multiplicity. Hybrid positive predictions occurred in 28/35 Barcelona and 26/31 Leverkusen matches; the largest match contributed 10.1% and 9.6%, respectively. Rule-arm recall was 6–17%, reflecting limited coverage.

5.2 Discovered Rules

The final Barcelona hybrid has four clauses (four sequential patterns) and Leverkusen three; each is effectively an OR over unary pattern atoms. Table V shows six representative rules of the 12 reviewed. B denotes Barcelona and L denotes Leverkusen. B6 is a coach-approved intermediate cross-validation candidate, not a clause in either final program; it was included among approved-rule indicators in the exploratory aggregate risk model. L4 is reviewed but not shown in Table V. Rates are in-sample and rules were partly selected for high rates, so they are optimistic (Fig. 2).

TABLE III. FIVE-METHOD THREE-FOLD OOF RESULTS AT NATURAL PREVALENCE

Team	Method	TP	Prec.	Recall	Lift (95% CI)	MCC
Barcelona	Hybrid	25	0.253	0.143	1.89 [1.48–2.33]	0.100
Barcelona	Miner only	16	0.271	0.091	2.03 [1.26–2.84]	0.088
Barcelona	CART	30	0.192	0.171	1.44 [0.95–1.93]	0.063
Barcelona	Popper, no patterns†	20	0.220	0.114	1.64 [1.20–2.13]	0.069
Barcelona	Context logistic‡	96	0.176	0.549	1.32 [1.09–1.54]	0.105
Leverkusen	Hybrid	17	0.205	0.096	1.26 [0.63–1.90]	0.033
Leverkusen	Miner only	10	0.222	0.056	1.36 [0.72–2.10]	0.033
Leverkusen	CART	14	0.255	0.079	1.56 [0.72–2.35]	0.057
Leverkusen	Popper, no patterns	27†	0.281†	0.153†	1.73† [1.09–2.32]	0.100†
Leverkusen	Context logistic‡	107	0.242	0.605	1.48 [1.31–1.67]	0.177

† Barcelona: pattern-free Popper found no program in fold 2 (zero positive predictions there); the row pools all three folds (TP=20, FP=71, FN=155, TN=1,062). Leverkusen: the same program was found in all folds (27 negative outcomes among 96 matching sequences). ‡ Features: n_actions, start_x, and start_y; threshold selection as in Section 4.4. OOF AUC: 0.591 (Barcelona), 0.645 (Leverkusen). Lift intervals are 95% match-cluster bootstrap intervals.

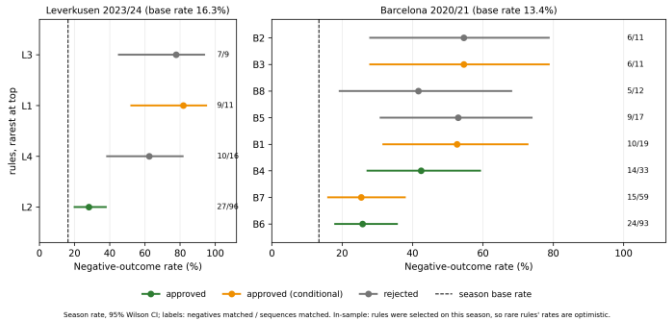


Fig. 2. Seasonal in-sample negative-outcome rates for the reviewed rules in Barcelona and Leverkusen. These selected-rule rates are descriptive and optimistic; they are not OOF arm-level lift estimates.

TABLE IV. PAIRED OUT-OF-FOLD LIFT DIFFERENCES

Team-season	Comparator	Hybrid − comparator lift	95% paired CI
Barcelona 2020/21	Miner only	−0.14	[−0.75, 0.45]
Barcelona 2020/21	Popper, no patterns	0.24	[−0.28, 0.78]
Barcelona 2020/21	CART	0.45	[−0.10, 1.05]
Barcelona 2020/21	Context logistic	0.57	[0.07, 1.10]
Leverkusen 2023/24	Miner only	−0.11	[−0.96, 0.74]
Leverkusen 2023/24	Popper, no patterns	−0.47	[−0.93, −0.003]
Leverkusen 2023/24	CART	−0.30	[−0.80, 0.21]
Leverkusen 2023/24	Context logistic	−0.23	[−0.95, 0.52]

Positive values favor the hybrid. Intervals use 20,000 shared match-cluster bootstrap draws; they are conditional on saved fits and not adjusted for multiple comparisons.

In Barcelona, Popper without patterns ended with NO SOLUTION in fold 2 without timing out (zero positive predictions) but found programs in folds 1 and 3, whereas the hybrid found a program in every fold. The pooled OOF row retains fold 2 as zero predicted positives. This suggests a lower search burden in these runs only; it is not a controlled estimate of the miner's general contribution because the arms differ in training population. In Leverkusen, the pattern-free arm found the same single-clause program in all folds, the hybrid retained the same atom, and all three CART trees had a root from the same family. This shared predicate covers both flanks without distinguishing left from right; these results do not establish a general weakness in backward flank carries.

Each step is predicate(action, channel L/C/R, third D/M); arrows indicate order with a maximum gap of two actions, while "+" denotes unordered sequence-level features. L4 shares its first two steps with L3. Rules with lifts around 3–5 have coverage of 33 or less and wide confidence intervals. More frequent rules (L2, B4, B6) are estimated more precisely but have lower lift. The apparent rarity–high-rate association is partly a selection effect.

5.3 Coach Review and Aggregate Risk Model

The coach approved five of eight Barcelona rules (three conditional) and two of four Leverkusen rules (one conditional), for 7 of 12 overall. Rejections mostly reflected limited evidence (few occurrences or zero to two holdout examples), not tactical incorrectness. The coach also retained observations from rejected rules: Leverkusen patterns were concentrated in the right channel, Barcelona patterns in the left. These are interpretations consistent with the patterns, not independent confirmation; rules can share steps and sequences (e.g., L3 and L4 share their first two steps). Further readings—high pressure, body orientation toward the line, or player identity—are expert interpretations.

TABLE V. REPRESENTATIVE RULES, IN-SAMPLE RATES (WILSON 95% CI), AND COACH DECISIONS

Rule	**Ordered pattern**	Support; rate (Wilson 95% CI) [27]	**Lift; decision**
L1	under_pressure(carry,R,D) → dead_end_support(carry,R,D)	9/11; 81.8% (52–95)	5.02; Conditional
L2	flank_backward_carry (single atom)	27/96; 28.1% (20–38)	1.73; Approved
L3	line_pass(pass,R,D) → connecting_pass(pass,R,D) → rest_defence_exposed(carry,R,D)	7/9; 77.8% (45–94)	4.77; Rejected
B1	rest_defence_exposed(pass,C,D) → rest_defence_exposed(pass,L,D) → under_pressure(carry,L,M)	10/19; 52.6% (32–73)	3.93; Conditional
B4	line_pass(pass,R,D) → first_time_play(pass,R,D)	14/33; 42.4% (27–59)	3.17; Approved
B6	line_pass + first_time_play + fast_sequence	24/93; 25.8% (18–36)	1.93; Approved

An aggregate risk model (L1 logistic regression over indicators for approved rules) discriminates weakly on the review holdout: AUC 0.566 (333 sequences) and 0.556 (273 sequences). Large odds ratios (e.g., 19.3 for Leverkusen L1, based on 11 occurrences) are unstable. This agrees with low recall. Because the holdout also informed candidate ranking, these AUCs are exploratory, selection-influenced estimates and not independent held-out evidence.

6. Discussion and Limitations

Predictive performance did not identify a stable winner across the two team-seasons: Barcelona favored the hybrid (lift 1.89 vs. 1.32) while Leverkusen reversed the ordering (1.26 vs. 1.48), a divergence consistent with differing 360 coverage (31 of 34 Leverkusen matches), team style, and base rate. The hybrid's distinct contribution is its compact, temporally grounded rules for targeted video review; the single coach's unblinded review offers preliminary face-validity evidence (7 of 12 rules approved, 4 conditionally), not comparative evidence of interpretability or utility.

Ordered motifs are explicit in the candidate representation, but this study does not isolate their empirical contribution. In Barcelona, the hybrid found a program in every fold, whereas pattern-free Popper found none in fold 2; because training populations differ, this does not establish a reduced search burden. CART shares the Prolog feature layer, so its comparison does not isolate Prolog's contribution. A mined pattern may also describe only the

conditions leading up to, not including, the event that breaks the sequence.

Limitations: (1) Each study covers one team in one season, so we make no generalization claim. (2) Table III uses match-cluster bootstrap intervals over fixed OOF predictions; these condition on the saved fold models and do not include model-refitting variability. With 31–35 match clusters, nominal 95% percentile-bootstrap coverage is uncertain and the intervals should be read as approximate. Only one three-fold outer split was evaluated, and paired lift comparisons are not adjusted for multiple comparisons. Wilson intervals for selected in-sample rules in Table V are sequence-level and not match-cluster adjusted. (3) Rules are selected from many candidates and rule rates are in-sample; a match-level permutation test was not run. We added a simple context baseline using sequence length and start coordinates, but other simple feature sets may perform differently. (4) The outcome is operationally defined; match-grouped CV reduces but does not remove dependence among sequences within a match. (5) 360 frames are incomplete and do not support off-ball concepts. Only 31 of Leverkusen's 34 available matches had usable 360 frames; this availability-based inclusion may limit representativeness. (6) One AFC-certified coach reviewed the rules without blinding; there is no inter-rater or blinded reliability estimate. (7) The discovery-fold test metrics used to rank intermediate candidates and the separate coach-review holdout scores are selection-influenced, so rule-level metrics and AUCs are exploratory. (8) Arms differ in training population (natural prevalence versus matched controls), so component contributions are not isolated. The system is therefore a hypothesis generator for video review, not a predictor.

7. Conclusion and Future Work

We present a hybrid pattern-mining and ILP method producing compact, auditable rules that combine ordered motifs with football predicates for targeted video review. Predictive lift varied by team-season, and comparisons establish no general winner. Rule arms have low recall, so the system aids hypothesis generation rather than comprehensive prediction; its value rests on auditable rules, most approved by an AFC-certified coach.

Future work: (1) Focused runs with coach-configured BK, such as goalkeeper-related predicates or a single region. (2) Reverse application to recent opponent matches to find fragile build-up patterns; whether inducing several rule conditions together increases opponent failure remains untested. (3) Full tracking and player identities may improve off-ball predicates, an expectation not yet tested; richer vocabularies also expand search and reduce support per rule. (4) Repeat match-clustered resampling with model refitting to quantify training variability, and controlled component ablations with identical training populations.

References

[1] Biermann, H., Wieland, F.-G., Timmer, J., Memmert, D., and Phatak, A. (2023). Towards expected counter—Using comprehensible features to predict counterattacks. Communications in Computer and Information Science, 1783, 3–13. doi:10.1007/978-3-031-27527-2_1

[2] Decroos, T., Bransen, L., Van Haaren, J., and Davis, J. (2019). Actions speak louder than goals: Valuing player actions in soccer. Proceedings of KDD, 1851–1861. doi:10.1145/3292500.3330758

[3] Singh, K. (2019). Introducing expected threat (xT). Blog post. https://karun.in/blog/expected-threat.html

[4] Van Roy, M., Robberechts, P., Decroos, T., and Davis, J. (2020). Valuing on-the-ball actions in soccer: A critical comparison of xT and VAEP. Proceedings of the AAAI-20 Workshop on AI in Team Sports, 1–8.

[5] Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, 1(5), 206–215. doi:10.1038/s42256-019-0048-x

[6] Cropper, A., and Dumančić, S. (2022). Inductive Logic Programming at 30: A New Introduction. Journal of Artificial Intelligence Research, 74, 765–850. doi:10.1613/JAIR.1.13507

[7] Cropper, A., and Morel, R. (2021). Learning programs by learning from failures. Machine Learning, 110(4), 801–856. doi:10.1007/s10994-020-05934-z

[8] Hocquette, C., Niskanen, A., Järvisalo, M., and Cropper, A. (2024). Learning MDL logic programs from noisy data. Proceedings of AAAI, 38(9), 10553–10561. doi:10.1609/aaai.v38i9.28925

[9] Breiman, L., Friedman, J. H., Olshen, R. A., and Stone, C. J. (1984). Classification and Regression Trees. Wadsworth.

[10] Decroos, T., Van Haaren, J., and Davis, J. (2018). Automatic discovery of tactics in spatio-temporal soccer match data. Proceedings of KDD, 223–232. doi:10.1145/3219819.3219832

[11] Dash, S., Ide, K., Umemoto, R., Amino, K., and Fujii, K. (2025). Prediction-based evaluation of back-four defense with spatial control in soccer. arXiv:2511.06191 (preprint).

[12] StatsBomb. (2021). StatsBomb announce the release of free StatsBomb 360 data: Euro 2020 available now. Blog post.

[13] Muggleton, S., and De Raedt, L. (1994). Inductive logic programming: Theory and methods. The Journal of Logic Programming, 19–20, 629–679.

[14] Gebser, M., Kaminski, R., Kaufmann, B., and Schaub, T. (2019). Multi-shot ASP solving with clingo. Theory and Practice of Logic Programming, 19(1), 27–82.

[15] Vercruyssen, V., De Raedt, L., and Davis, J. (2016). Qualitative spatial reasoning for soccer pass prediction. Proceedings of the 3rd Workshop on Machine Learning and Data Mining for Sports Analytics, CEUR Workshop Proceedings, vol. 1842 (ECML/PKDD 2016).

[16] Van Haaren, J., Dzyuba, V., Hannosset, S., and Davis, J. (2015). Automatically discovering offensive patterns in soccer match data. Advances in Intelligent Data Analysis XIV, LNCS 9385, 286–297. doi:10.1007/978-3-319-24465-5_25

[17] Jacobs, N., Driessens, K., and De Raedt, L. (1998). Using ILP-systems for verification and validation of multi-agent systems. In Inductive Logic Programming, LNCS 1446. Springer. doi:10.1007/BFb0027318

[18] Dong, G., and Li, J. (1999). Efficient mining of emerging patterns: Discovering trends and differences. Proceedings of KDD, 43–52.

[19] Pei, J., Han, J., Mortazavi-Asl, B., Pinto, H., Chen, Q., Dayal, U., and Hsu, M.-C. (2001). PrefixSpan: Mining sequential patterns efficiently by prefix-projected pattern growth. Proceedings of ICDE, 215–224.

[20] Antunes, C., and Oliveira, A. L. (2003). Generalization of pattern-growth methods for sequential pattern mining with gap constraints. In Machine Learning and Data Mining in Pattern Recognition, LNCS 2734. Springer. doi:10.1007/3-540-45065-3_21

[21] Kapoor, S., and Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. Patterns, 4(9), 100804. doi:10.1016/j.patter.2023.100804

[22] Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography, 40(8), 913–929. doi:10.1111/ecog.02881

[23] Wielemaker, J., Schrijvers, T., Triska, M., and Lager, T. (2012). SWI-Prolog. Theory and Practice of Logic Programming, 12(1–2), 67–96. doi:10.1017/S1471068411000494

[24] Kramer, S., Lavrač, N., and Flach, P. (2001). Propositionalization approaches to relational data mining. In Relational Data Mining, 262–291. Springer.

[25] Li, C., Yang, Q., Wang, J., and Li, M. (2012). Efficient mining of gap-constrained subsequences and its various applications. ACM Transactions on Knowledge Discovery from Data, 6(1). doi:10.1145/2133360.2133362

[26] Field, C. A., and Welsh, A. H. (2007). Bootstrapping clustered data. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 69(3), 369–390. doi:10.1111/j.1467-9868.2007.00593.x

[27] Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association, 22(158), 209–212. doi:10.1080/01621459.1927.10502953